Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Static friction

Meaning in my words How would you explain static friction to a friend?	**How can you tell?** You push a crate gently and it stays still. What force balances your sideways push?
Example or sketch A stationary crate on a level floor is pushed right with 3 N below the sliding threshold. Draw friction.	**Non-example and why** The crate is already sliding across the floor. Would the friction at that contact still be static?

Maximum static friction

Meaning in my words How would you explain maximum static friction to a friend?	**How can you tell?** The floor can supply up to 6 N. How much friction balances a gentle 2 N horizontal push?
Example or sketch With a 6 N limit and no other horizontal forces, what do you predict for a 7 N push?	**Non-example and why** A friend writes 6 N for every trial, even with no sideways push. What does 6 N mean?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Kinetic friction

Meaning in my words How would you explain kinetic friction to a friend?	**How can you tell?** A crate slides right across a stationary floor. Which way does kinetic friction act on the crate?
Example or sketch If the coefficient is 0.2 and the normal force is 10 N, what arrow would you draw?	**Non-example and why** A crate stays put while you push gently. Should you use the kinetic-friction rule?

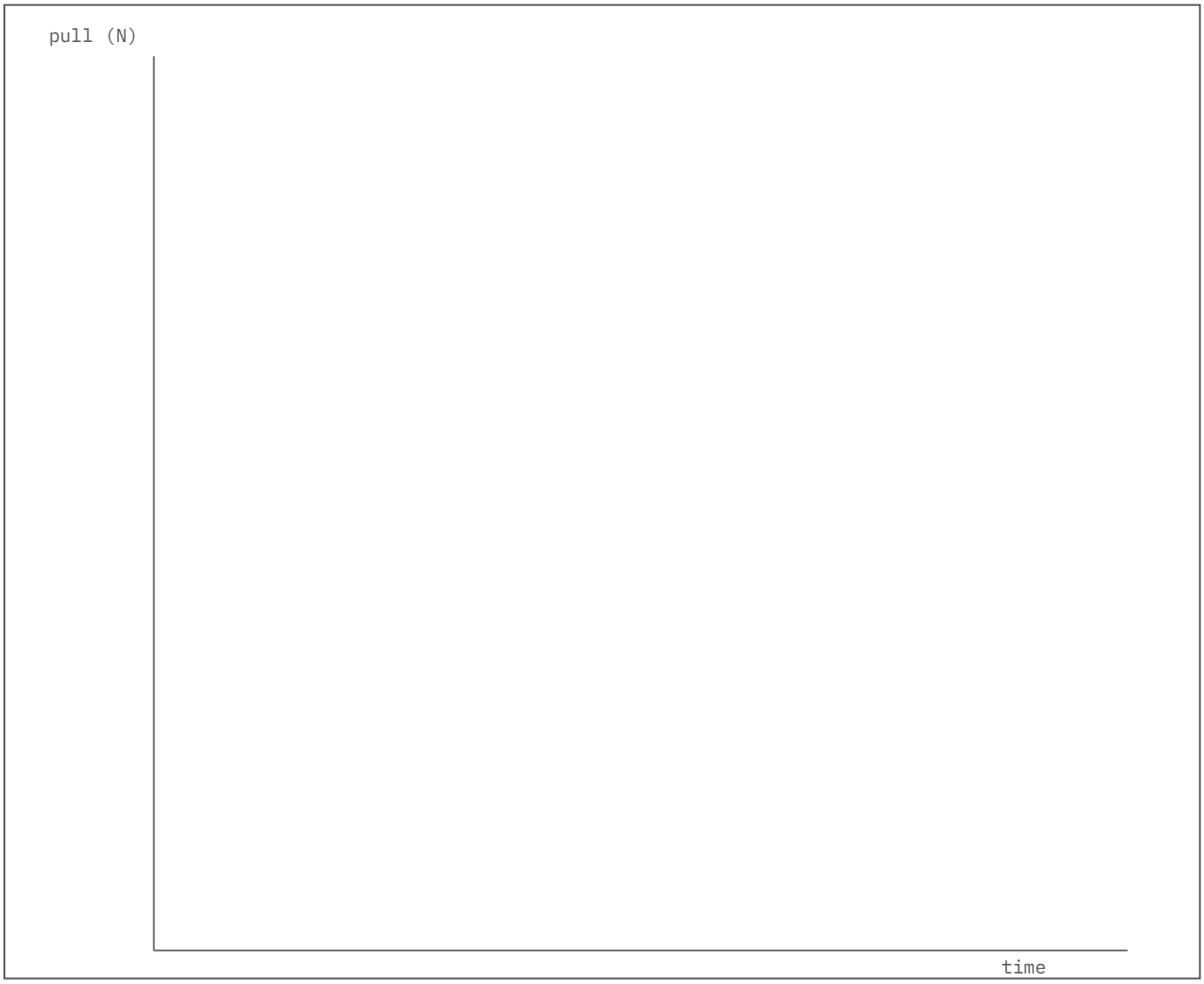
Coefficient (μ)

Meaning in my words What does the coefficient of friction tell us about two surfaces?	**How can you tell?** A sliding block has 2 N friction and 10 N normal force. How do you find the kinetic coefficient?
Example or sketch With the same surfaces and a 20 N normal force, what sliding friction would μ = 0.2 predict?	**Non-example and why** A student writes “μ = 0.2 N.” What happens to the units in the ratio?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.



My marks mean

Cut from the table to space

17

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use contact surfaces to decide when friction acts.

CHOOSE ONE JOB TOGETHER

EFFECTS ENGINEER ☐

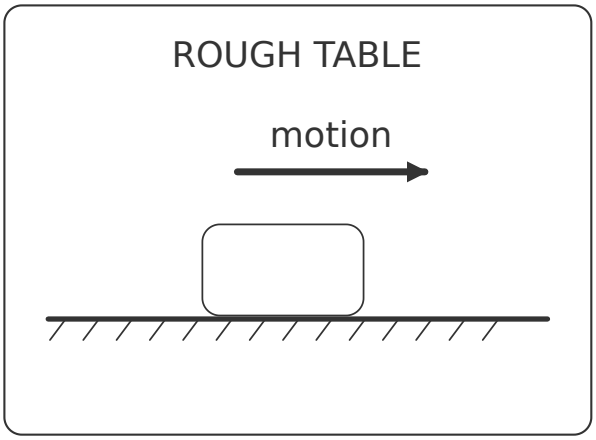
Set friction for each scene. Mark the touching surfaces and explain what could change XJ's motion in empty space.

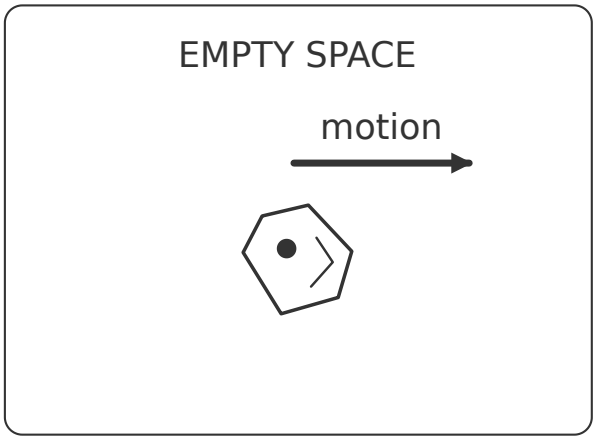
SCIENCE BROADCASTER ☐

Correct the caption for the two shots. Point to the surfaces and say what could still bend XJ's path.

Point to a clue. Keep your first marks; add the update using the color key.

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says “Both will stop from surface friction.”





01 OUR FIRST CALL

What should the two-scene caption say? Explain using the contact surfaces.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the new shot? Update the force explanation for XJ.

Switch speakers. Which surfaces are touching?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks